Predicting Candidate Biomarkers for COVID-19 associated with Leukemia in Children

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INTRODUCTION

- Children have shown increased immunity to COVID-19 (Mallapaty, 2021)
- Acute Myeloid Leukemia (AML) is one of the most common types of acute leukemia in adults but is rare in children
- Leukemia developed in children after recent infection of COVID-19 (Ioannidou et al., 2021)
- *IFIH1* encodes MDA5, a sensor that activates the immune response
- MDA5 is responsible for children's increased immunity to COVID-19 (Fischer, 2021)
- Previous studies found that children have a higher expression of *IFIH1* compared to adults
- MicroRNAs (miRNAs), or small (~ 17-22bp) non-coding RNAs, regulate the gene expression
- **My project goal is to identify candidate genes, in the context of miRNA targeting mechanism, that could serve as biomarkers to coronavirus disease and leukemia in children.**

OBJECTIVES & HYPOTHESIS

The objectives of my research are: 1) Is *IFIH1* also a genetic biomarker linked to AML? 2) Are there any additional candidate genes associated with COVID-19 and AML?

Hypothesis: *IFIH1* and its regulator miRNAs are biomarkers linked to AML cancer in children

DATABASES AND METHODS

- I wanted to use *IFIH1* as the reference gene in order to find the COVID-19 virus's connection to AML in children
- First, I needed to prove that *IFIH1* was indeed an important and significant gene that will lead me to the connection between COVID-19 and AML
- TarBase, TCGA, and STRING were used to prove IFIH1's significance
- After *IFIH1* was confirmed as significant, I analyzed the data I acquired from those steps to prove COVID-19's connection to AML in children.
- GEPIA, UCSC Xena, DAVID, and ImmGen were used to analyze my acquired data

Bioinformatics Tools

TarBase

Searched for i) targeting miRNAs by the *IFIH1* gene, and ii) additional genes targeted by miRNAs that target the *IFIH1* gene.

TCGA

Retrieved expression information for the targeted genes from RNA-Seq data.

STRING

Identified Protein-Protein Interaction gene set for *IFIH1* gene.

GEPIA
Gene Expression Profiling Interactive Analysis

Expression in cancer vs. normal

DAVID BIOINFORMATICS DATABASE

Obtained common terms/pathways.

UCSC Xena

Expression correlation between candidate genes & density of expression for samples

ImmGen

Checked for any experimental evidence for the candidate genes' expressions in immune cells

RESULTS

IFIH1 Expression is linked to Viral Infection and Acute Myeloid Leukemia

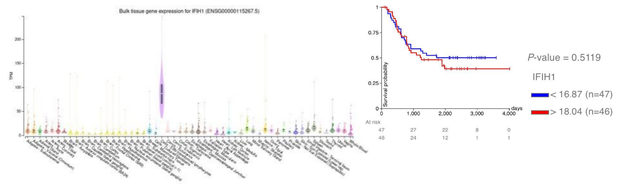


Figure 1.

GTEx results show IFIH1 has a high expression in EBV-transformed lymphocytes, which is important for killing tumor cells and creating antibodies for the immune system.

Higher expression of IFIH1 decreases patient's survival rate in Acute Myeloid Leukemia.

miRNAs and Genes Interacting with IFIH1

Gene name	miRNA name	Methods	Pred.Score
IFIH1 (hsa)	hsa-miR-196a-5p	IP	0.753
IFIH1 (hsa)	hsa-miR-196b-5p	IP	0.752

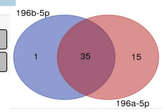


Figure 2.

Found two targeting miRNAs of IFIH1 after searching TarBase; hsa-miR-196a-5p and hsa-miR-196b-5p.

Then use the same miRNAs to identify significant genes targeted by the miRNAs; obtained approximately 150-170 genes; use Unix putty commands to select genes with >500 TPM.

Obtained 51 candidate genes targeted by both miRNAs.

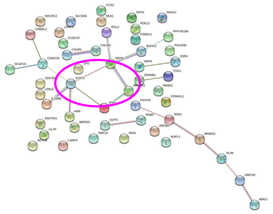


Figure 3.

Conducted PPI using STRING on the 51 candidate genes acquired from previous step.

Found that there are two genes, STAT3 and MAP3K1, that directly interact with IFIH1

Activated STAT3 can down-regulate lymphopoiesis and impair the anti-virus immune responses (Jafarzadeh et al., 2021).

MAP3K1 can facilitate APL (the M3 subtype of AML) cell proliferation (Tang et al., 2021).

Candidate Gene's Expressions in TCGA Cancers, GTEx Normals, and UCSC Xena Trials

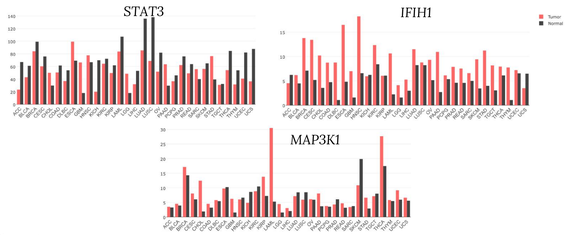


Figure 4.

The three candidate genes tend to have a much higher expression in TCGA cancer samples compared to GTEx normal samples, indicating that they have a role in cancer development.

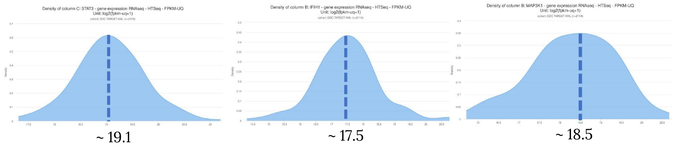


Figure 5.

Majority of AML Samples from TARGET-AML Trial have High Expression of IFIH1, STAT3, and MAP3K1

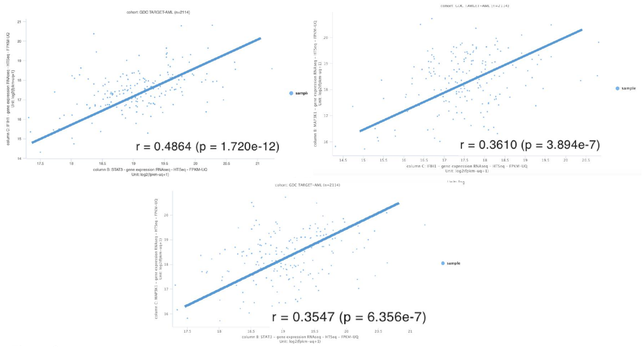


Figure 6.

Correlation of STAT3, MAP3K1 and IFIH1 Expressions in TARGET-AML Samples.

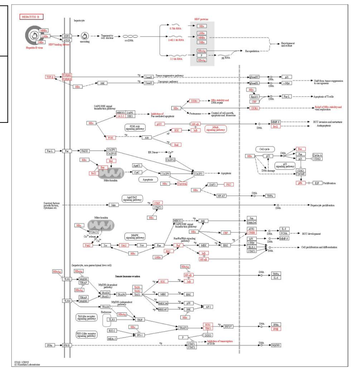
Results show that all three candidate genes have a positive correlation with each other, thus meaning that their expressions are related.

IFIH1, *STAT3*, and *MAP3K1* are Found in Another Cancer Pathway

Term	Genes	P-Value	Fold-Enrichment
hsa05161:Hepatitis B	*IFIH1, MAP3K, STAT3*	4.41E-4	47.44

Figure 7.

DAVID and Kegg Pathway results show that the three candidate genes are also involved in the Hepatitis B (p-value < 0.0004).



IFIH1 and *STAT3* Expression Data in Mice

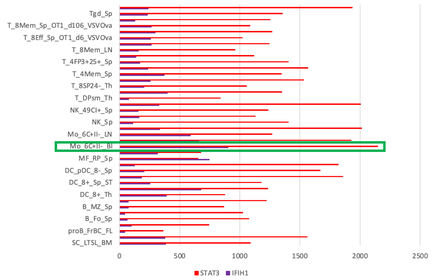


Figure 8.

Inputted both IFIH1 and STAT3 into ImmGen.

Results show that the mouse orthologous genes of STAT3 and IFIH1 are co-expressed high in CD16+ monocytes.

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CONCLUSIONS & FUTURE PLANS

- *IFIH1* is likely a candidate biomarker for AML.
- Two genes (STAT3 and MAP3K1) were identified to have a direct interaction with *IFIH1*.
- All three genes could contribute to the comorbidity of COVID-19, AML, and HBV.
- Four additional genes from ImmGen's gene constellation graphs were seen as biomarkers for the comorbidity of complex diseases.
- Future studies:
  - Study the function of *IFIH1* expression in viral infection and leukemia development
  - Develop a machine-learning model to predict more candidate genes as biomarkers for leukemia

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